

AI Career Intelligence Platform

Software Architecture Document

Software Design Document (SDD)

High-Level Design (HLD)

Low-Level Design (LLD)

Architecture Decisions (ADR)

Technology Stack

Version: 1.0

Prepared By: Akshay Mohan

Table of Contents

- Introduction
- Software Design
- High-Level Design
- Low-Level Design
- Architecture Decisions
- Technology Stack
- Conclusion

Introduction

The AI Career Intelligence Platform helps users analyze resumes, extract skills, recommend jobs, predict salaries, generate learning roadmaps, and interact with an AI-powered career assistant.

Software Design

Layer	Description
Frontend	Next.js
Backend	FastAPI
Database	PostgreSQL
Cache	Redis
AI	spaCy, scikit-learn, LangChain

High-Level Design

Architecture Flow:

User → Next.js → FastAPI → PostgreSQL / Redis → AI Services → Response

Low-Level Design

Module	Responsibility
Authentication	JWT Login & Registration
Resume	Upload & Parsing
Jobs	Matching Engine
Salary	Prediction
Learning	Roadmap
Chat	AI Assistant

Architecture Decisions

- FastAPI for backend
- Next.js for frontend
- PostgreSQL for structured data
- Redis for caching
- Docker for deployment
- JWT for authentication

Technology Stack

Category	Technology
Frontend	Next.js, React
Backend	FastAPI
Database	PostgreSQL
AI/ML	spaCy, scikit-learn, LangChain
DevOps	Docker, GitHub
Security	JWT, bcrypt, HTTPS

Conclusion

This document summarizes the architecture and design decisions for Version 1.0 of the AI Career Intelligence Platform.